MILITARY SPECIFICATION

DEGREASER, PORTABLE, SOLVENT TYPE, TANK IMMERSION

This specification is approved for use within the US Army Armament Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable, solvent type degreaser capable of cleaning parts and objects by fine spraying, heavy spraying, full volume flow or minimal flow washing, or by immersion.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

P-D-680	-	Dry Cleaning Solvent.
W-C-596	-	Connector, Electrical, Power, General Specification For.
W-C-596/13	-	Connector, Plug, Electrical, General Purpose, Hospital Grade, Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts, 50/60 Hertz.
CC-M-1807	-	Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-T-60	-	Tape, Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-T-152 - Treatment, Moisture- and Fungus-Resistant. Of Communications, Electronic, and Assorted Electrical Equipment.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.
ASTM E380-84 - Standard for Metric Practice.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Control and Systems.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.
NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The degreaser shall be a portable type designed to pump dry cleaning solvent (see 3.8.2) through a flexible hose for the purpose of cleaning parts and objects by circulation, fine spraying, heavy spraying, full volume flow or minimal flow washing, or by immersion. The degreaser shall be constructed in a manner similar to the illustration shown in Figure 1. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the degreaser or any component.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the degreaser. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

3.2.2 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a

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particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.3 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.4 Safety and health requirements. Covers, guards, and other safety devices shall be provided. The devices shall not interfere with operation of the degreaser. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of parts. All parts, components, mechanisms, and assemblies furnished on the degreaser shall comply with the requirements of OSHA 2206 that are applicable to the degreaser itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The degreaser shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation at maximum rating and capacity without distortion. The degreaser shall be furnished complete so that, when installed and connected to a proper power source, it can be used for the operations specified herein.

3.4.1 Castings and forgings. Castings shall be free from visible blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from visible scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to renewal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

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3.4.4 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.5). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the degreaser to remove the temporary caps or stoppers before start-up or use of the degreaser.

3.4.4.1 Lubrication chart or plate. A corrosion-resistant metallic nameplate, with permanently legible characters written in the English language, shall be attached to each degreaser in a location visible to the operator. If a lubrication chart is furnished, it shall be placed in a transparent plastic folder or laminated between clear, permanently sealed, plastic sheets with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application
Service interval
Type of lubricant
Viscosity
Military or Federal specification number of lubricant

3.4.5 Surfaces. Surfaces of castings, forgings, molded parts, stampings, and welded parts shall be clean and free of sand, dirt, fins, sprues, scale, flux, flash, and other harmful or extraneous materials.

3.4.6 Protective finishes. Exposed metal and working surfaces, except when the parts involved are inherently corrosion resistant, shall be protected with an electrochemical or chemical conversion finish (see 6.6).

3.4.7 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured, and soldered or brazed. Only rosin base fluxes shall be used in soldering operations.

3.5 Performance and product characteristics.

3.5.1 Load and shock. The degreaser shall be constructed to be moved manually across the floor in any direction, while loaded with water or solvent (2/3 tank capacity) and a minimum of 250 pounds of parts. The degreaser shall be capable of traversing a one inch depression and moving up and over a 15 degree slope, one inch high, at a minimum travel rate of 25 feet per minute while loaded to the capacity specified above (see Figure 2). There shall be no evidence of damage, deformation, breakage, or leakage.

3.6 Components.

3.6.1 Tank and hinged cover. The tank shall have the following minimum inside dimensions: 102 inches long, 10 inches wide, and 12 inches deep. The tank

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and cover shall be fabricated from sheet steel, not less than Manufacturers' Standard Gage No. 16 (0.060 inch), galvanized on the inside surface. The cover shall be hinged along the length dimension (102 inches) and shall be furnished with a fusible link which shall close the cover automatically in the event of fire. The construction of the tank (cover excluded) shall assure a liquid tight container.

3.6.1.1 Drain valve. A drain valve shall be installed in the bottom of the tank to permit complete drainage of the tank without using the pump to drain the tank.

3.6.1.2 Strainer. A strainer shall be installed between the tank and pump inlet hose to prevent any foreign matter, that would impede operation, from entering the pump. The strainer shall be constructed to permit removal and replacement without requiring draining the solvent from the tank.

3.6.2 Pump. The pump shall be resistant to cleaning solvents, shall be either the centrifugal vane type or the turbine type, and shall be on drive centerline with the motor shaft. The pump shall be capable of delivering solvent, through the hose with control valve (nozzle) attached, at a minimum rate of 225 gallons per hour for a minimum period of eight hours.

3.6.2.1 Pump inlet hose. The inlet hose shall be solvent resistant, and shall extend from the strainer, at the bottom of the tank, to the inlet side of the pump.

3.6.2.2 Discharge hose and nozzle. The discharge hose shall be solvent resistant and shall be a minimum of 5 1/2 feet long with a 3/8 inch inside diameter. The hose shall be furnished with an adjustable nozzle to permit either a fine or heavy spray, and to regulate from a full volume solid stream down to a minimal stream. The discharge hose shall be located to provide accessibility to the full working area of the tank. A means shall be provided to permit hanging the nozzle along the side of the tank just below the top edge. The hose shall be located to prevent inadvertent drainage of the tanks.

3.6.3 Frame and base. The height of the degreaser, from floor level to the top of the tank, shall be 35 inches $\pm$ one inch. The frame and base shall support the tank and other components such as the pump, motor, wiring, and piping. The frame and base shall be constructed of metal to support and withstand the load and shock requirements 3.5.1.

3.6.3.1 Casters. The base shall be mounted on four, antifriction bearing type, casters having a minimum of three inches in diameter, with one inch tread width. The casters shall be the swivel type, permitting movement of the degreaser in any direction, with locking devices capable of preventing movement when required.

3.6.4 Work basket. The work basket shall be constructed of perforated stainless steel, Manufacturers' Standard Gage No. 16, with full frame top and bottom. The basket shall be 24 inches long, 8 inches wide, and 4 inches deep. The work basket shall include two hinged handles constructed to permit folding against the side of the basket when not in use.

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3.6.5 Work rests. Two work rests shall be provided with each degreaser for the purpose of supporting the parts being cleaned and preventing the parts from resting on the bottom of the tank. One work rest shall be 36 inches long by 9 3/4 inches wide, with flanges 2 1/2 inches high along the 36 inch dimension. The other work rest shall be 29 inches long by 9 3/4 inches wide, with flanges 2 1/2 inches high along the 29 inch dimension. The work rests shall be fabricated from expanded galvanized metal, and shall be capable of supporting a minimum load of 250 pounds.

3.6.6 Electrical equipment.

3.6.6.1 Electrical system. The electrical system shall conform to applicable requirements of NFPA 79. A disconnect switch shall be provided to disengage all incoming power, and shall be separate from the motor's on and off switch. The motor shall be capable of operating on 115 volt, AC, single phase, 60 Hertz power and shall be initially wired to operate on 115 volts. An identified terminal suitable for connecting the proper size grounding conductor in accordance with the requirements for the specified power source shall be provided.

3.6.6.1.1 Motor. The motor shall be an explosion proof, squirrel-cage, induction type, of no less than 0.250 horsepower conforming to CC-M-1807, with class B insulation and a continuous duty rating.

3.6.6.1.2 Switch, foot operated. The pump motor shall be turned off and on by a two pole, explosion proof, foot switch.

3.6.6.2 Electrical components. All electrical apparatus components, including but not limited to relays, switches, motors, wiring, and their inclosures, shall be explosion proof in accordance with NEMA Electrical Standards (Type 7, Class 1, Group D). In addition, all electrical components shall conform to, and be located in accordance with, NEMA ICS 1 and NFPA 70, as applicable. No component shall be used in any application where the requirements exceed the manufacturer's rating for the part.

3.6.6.3 Electrical power supply cable. An electric power supply cable shall be installed on each degreaser. The electrical power supply cable shall be not less than ten feet in length, and shall terminate in a three prong grounding plug in accordance with W-C-596/13 and NFPA 70 (National Electric Code). The three prong plug shall be furnished with a three to two prong outlet adapter with a grounding lead. The electric power supply cable shall include sheathing/covering in accordance with NFPA 70 and OSHA 2206. The cable shall have a voltage/potential and amphere/current carrying capacity exceeding that required by the electric power consuming devices, including electric motors, of the degreaser in accordance with NFPA 70.

3.6.6.4 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the

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industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.8 Product identification.

3.8.1 Nameplate. The nameplate and information plates may be incorporated into one or more corrosion-resistant metallic plates at the manufacturer's option. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The plates shall be secured to the degreaser with permanent fasteners such as screws, bolts, and rivets at each of the four corners. The nameplate shall include the following information:

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)
Contract number

3.8.1.1 Information plate. A corrosion-resistant metallic information plate shall be permanently fastened to the degreaser in a location visible to the operator. The information plate shall include warnings, cautions, and machine operator instructions to be followed when operating or maintaining the degreaser.

3.8.2 Warning plate. A warning plate made of corrosion-resistant metallic material shall be securely attached to the degreaser in a location where it shall be in full view. The wording shall be in the English language using permanent bold faced letters on a contrasting background. The letters shall be not less than 3/4 inch high for the word "WARNING" and 1/4 inch high for all other information. The warning notice shall be as follows:

WARNING
USE ONLY SOLVENTS CONFORMING TO SPECIFICATION P-D-680.
P-D-680 TYPE I, FLASH POINT 100°F, NSN 6850-00-285-8012 (55 GAL. DRUM).
P-D-680 TYPE II, FLASH POINT 138°F, NSN 6850-00-285-8011 (55 GAL. DRUM).
THIS DEGREASER, WHEN USED IN A CONFINED AREA, SHALL BE PROVIDED WITH A FORCED VENTILATION SYSTEM.

3.9 Workmanship. Standards of workmanship shall assure that the degreaser shall have the stability, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of a through c below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Performance test (see 4.6.1).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Product examination	II	2.0
Performance test	II	1.0
Packaging inspection	S-4	4.0

4.5 Product Examination. Visually and manually examine the degreaser to determine conformance with the requirements of 3.2 through 3.4.7 and 3.6 through 3.9. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which

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it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the degreaser has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Performance test. Fill the degreaser to the recommended operating level and start the pump motor. The unit shall be operated for a period of ten minutes in each cleaning mode specified in 3.2. Failure of the pump to operate, or any leakage of the unit, shall constitute failure of the test.

4.6.2 Pump capacity test. With the tank half full of solvent or water, the nozzle shall be fully opened and the pump turned on. The fluid shall be discharged to allow time for complete priming of pump and displacement of air in the system. After purging, the pump shall discharge the test fluid at a minimum rate of 225 gallons per hour for a minimum period of eight hours (see 3.6.2).

4.6.3 Nozzle test. While the pump is running, the nozzle shall be manipulated to determine spray and stream performance (see 3.6.2.2).

4.6.4 Load and shock test. The degreaser tank shall be loaded with water or solvent (2/3 tank capacity) and weighted with a minimum of 250 pounds of parts. The load and shock test shall be conducted in accordance with the method specified in Figure 2. The degreaser shall be traversed across a one inch depression (Obstacle No. 1) at a 45 degree approach angle and a minimum travel rate of 25 feet per minute. The movement shall be such that one caster at a time traverses the depression. The degreaser shall then be moved up and over a 15 degree slope, one inch high, (Obstacle No. 2) in the same direction and at a travel rate of 25 feet per minute. Obstacles No. 1 and 2 shall be of hard wood, concrete, or steel. Damage, deformation, breakage, or leakage shall be cause for rejection (see 3.5.1).

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Disassembly. The degreaser may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly, when applicable, shall be as specified in MIL-STD-1186.

5.2 Preservation. Preservation shall be Level A or Industrial, as specified in the contract (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The degreaser shall be cleaned and dried in accordance with MIL-P-116. Cleaning shall be in accordance with process C-1.

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5.2.1.2 Preservative application. Immediately after cleaning and drying, preserve all exterior surfaces subject to corrosion, including those exposed by disassembly, with type P-1 preservative conforming to MIL-P-116. Apply type P-3 preservative to the interior surfaces by flushing or pumping through the system. After draining, all openings shall be sealed with tape conforming to PPP-T-60, type optional, or caps or plugs conforming to MIL-C-5501. Caster bearings shall be preserved with type P-11 preservative.

5.2.1.3 Unit pack. The degreaser shall be individually packed in accordance with method I of MIL-P-116.

5.2.1.3.1 Casters. If casters are detached from the basic unit during the disassembly process, they shall be packed in a snug fitting fiberboard box conforming to PPP-B-636, class domestic.

5.2.1.3.2 Electrical equipment. The electric motor shall be preserved, using the alternate method specified for assembled machines, in accordance with Level A requirements of MIL-E-16298. The power cable shall be coiled to a minimum safe diameter and then secured to prevent uncoiling. The electrical plug and exposed ends of the power cable shall be covered with tape conforming to PPP-T-60, type IV. Openings in the electrical switch and outlet box shall be sealed with the same tape.

5.2.1.3.3 Discharge hose and nozzle. The hose shall be coiled to a minimum safe diameter and the coil secured with cotton tape or twine. Pack the hose and nozzle in a fiberboard box as specified in 5.2.1.3.1.

5.2.1.3.4 Technical publications. Publications and instructions, when applicable, shall be packed in accordance with method IC-1 of MIL-P-116.

5.2.1.3.5 Consolidated packaging. All packed component parts, accessories, and publications shall be adequately cushioned and packed in one or more containers conforming to PPP-B-636, W5c. Containers shall be closed in accordance with the container specification or the appendix thereto. The packed boxes shall be placed inside the tank and secured in a manner to restrict movement.

5.2.2 Industrial. Industrial preservation shall be as specified in ASTM D3951-82.

5.3 Packing. Packing shall be Level A, Level B, or Industrial, as specified in the contract (see 6.2.1).

5.3.1 Level A. All components and accessories comprising one complete degreaser, with technical publications when applicable, shall be packed in a container conforming to PPP-B-621, class 2. Blocking, bracing, anchoring, and cushioning shall be in accordance with the container specification and MIL-STD-1186. Containers shall be closed and strapped in accordance with the container specification and the appendix thereto.

5.3.2 Level B. The degreaser shall be packed as specified in 5.3.1 except it shall be packed in a container conforming to PPP-B-621, class 1.

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5.3.3 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

5.4 Marking. Marking shall be Military or Industrial, as specified in the contract (see 6.2.1).

5.4.1 Military. Marking shall be as specified in MIL-STD-129.

5.4.2 Industrial. Marking shall be as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. The portable degreaser covered by this specification is intended to be used for cleaning components and parts by immersion, spraying, and washing.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Exceptions and additional safety/health requirements when required (see 3.2.4).
(c) Painting and finishing if different (see 3.7).
(d) First article inspection if different (see 4.3).
(e) Quality conformance inspection if different (see 4.4).
(f) Preservation, packing, and marking as required (see 5.2, 5.3, and 5.4).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be identified as shown in MIL-M-7298.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be as specified by the user.

6.5 Standard lubricants. The degreaser manufacturer should specify and use to the maximum extent possible those lubricants (see 3.4.4) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

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6.6 Definitions.

6.6.1 Exposed metal. Exposed metal is defined as metal exposed to atmospheric conditions, solvents, oils and greases, and cleaning chemicals and fluids (see 3.4.6).

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - AL

Review activities:

Army - MD, SM

Preparing activity:

Army - AL

Project No. 4940-A486

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from the instrument to the dynamometer junction panel thermocouple

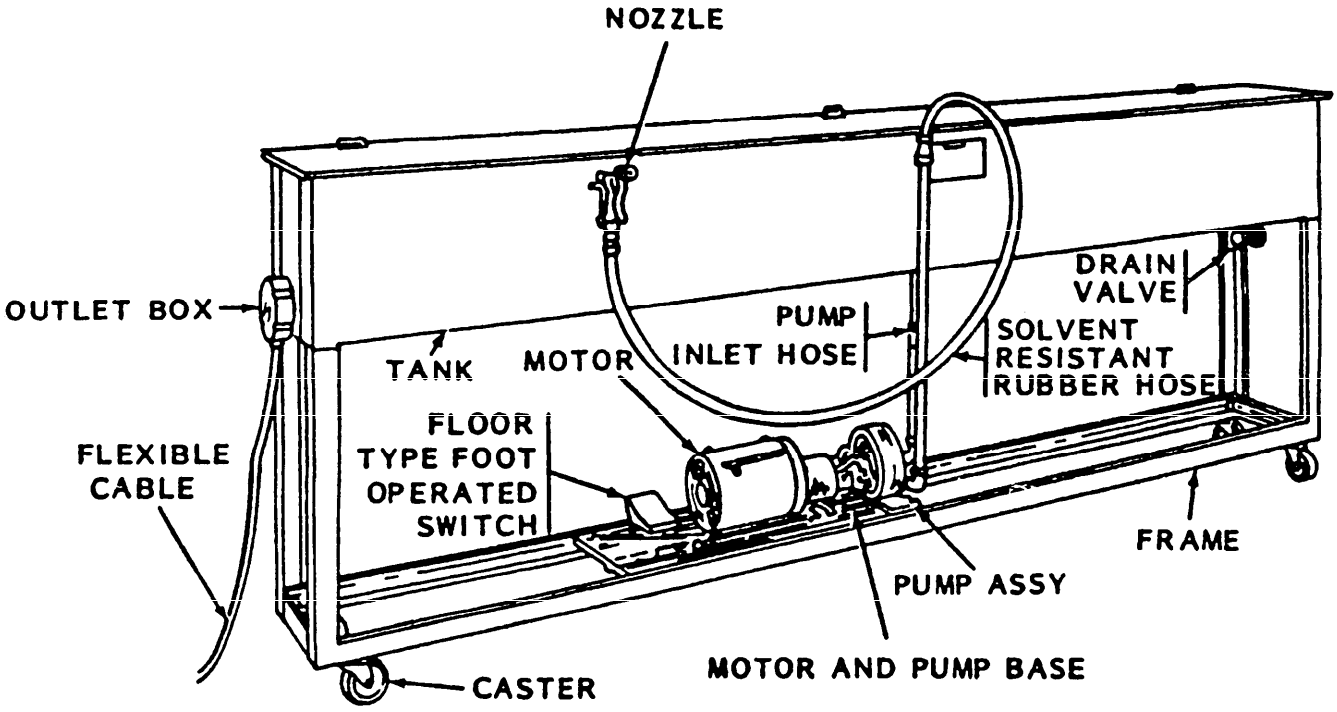


FIGURE 1. Degreaser.

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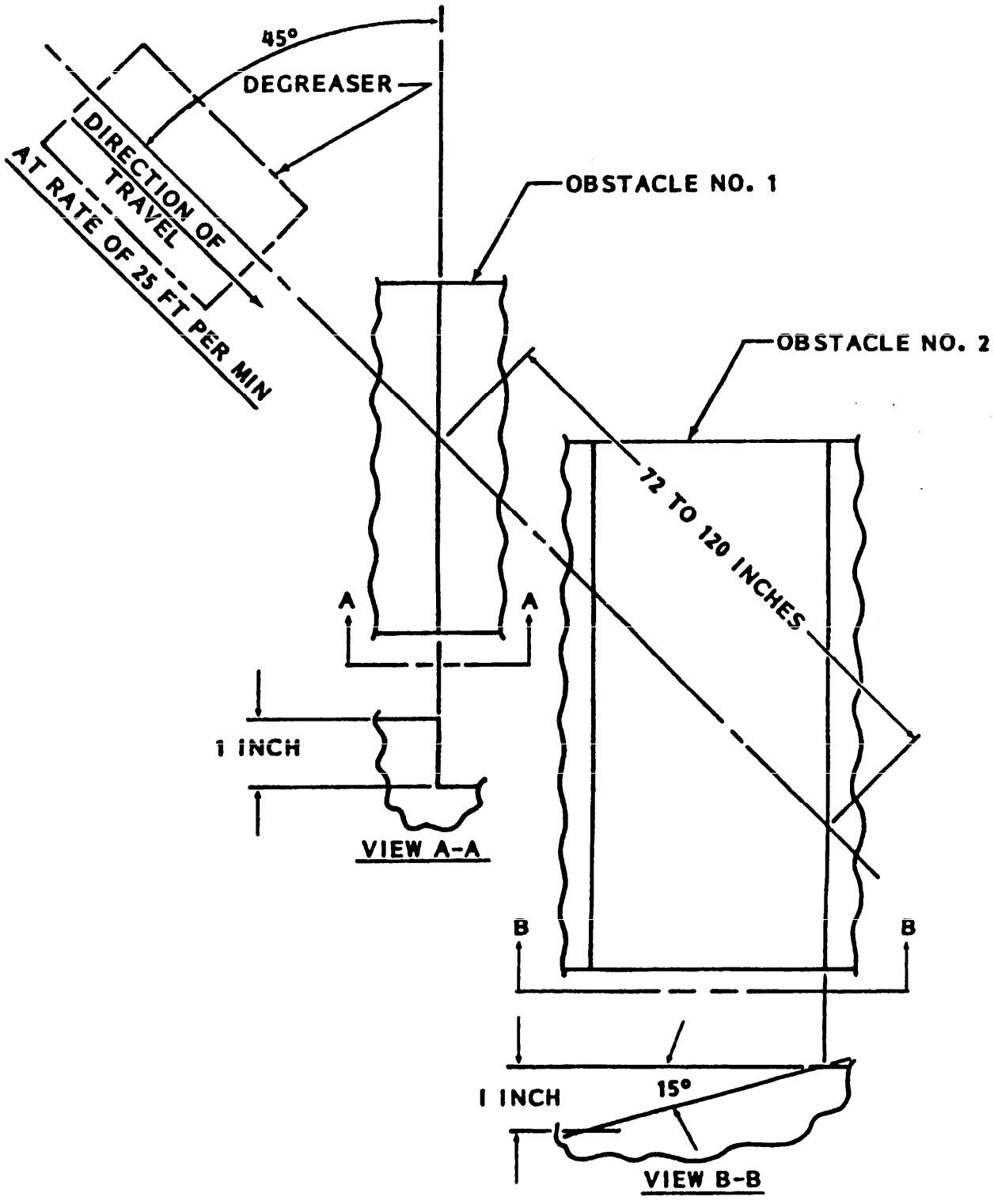


FIGURE 2. Load and shock test method.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-45064D	Degreaser, Portable, Solvent Type, Tank Immersion

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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